

Image Processing & Recognition: Lecture 1 Review

Welcome to your study guide! This document is designed to decode the first lecture of your "Image Processing and Recognition" course. We will move from the theoretical concepts to the practical math of training a simple AI model.

Part 1: What are you learning? (Course Breakdown)

The PDF you uploaded is an introductory lecture that builds the mathematical foundation for modern Computer Vision and Deep Learning. Here is the "Human-Readable" summary of the key concepts:

1. The Hierarchy of AI

You learned that **Deep Learning** is a subset of **Machine Learning**, which is a subset of **Artificial Intelligence**.

- **Traditional Programming:** You give the computer rules + data → it gives answers.
- **Machine Learning:** You give the computer data + answers → it figures out the rules (the model).

2. Linear Regression (The "Hello World" of ML)

The core of this lecture is **Linear Regression**. This is simply finding the best-fitting straight line through a set of data points.

- **The Model:** $\hat{y} = w_1x + w_0$. (Prediction = Weight $\times$ Input + Bias).
- **The Goal:** Find the specific values for w_1 and w_0 that make the error (difference between your line and the real dots) as small as possible.
- **The Solution:** How do we find those w 's?
 1. **Closed Form (Normal Equation):** A direct matrix formula that gives the exact answer instantly. (This is what Exercise 1 uses).
 2. **Gradient Descent:** An iterative method where you guess the weights, check the error, and nudge them slightly in the correct direction, repeating until the error is low.

3. Advanced Concepts

The lecture also briefly touched on:

- **Polynomial Regression:** When a straight line isn't enough (curved data), we add powers (x^2, x^3) to fit curves.
- **Regularization (Ridge/Lasso):** Techniques to stop the model from memorizing the data too perfectly (overfitting) by penalizing large weights.
- **Classification:** Using similar math to predict categories (e.g., "Cat" vs. "Dog") using **Logistic Regression** (binary) or **Softmax Regression** (multi-class).

Part 2: The Solved Problem (Step-by-Step)

Let's look at the exercise from the end of your PDF (Page 91).

The Task: You have 3 data points: $(1, 2)$, $(2, 4)$, $(3, 5)$. Find the best line $\hat{y} = w_1x + w_0$ that fits these points.

The Strategy: We will use the **Normal Equation**: $w^* = (X^T X)^{-1} X^T y$.

Step 1: Create the Matrices

First, we separate the data into inputs (X) and outputs (y).

- y is easy: it's just the column of target values $[2, 4, 5]^T$.
- X is the "Design Matrix". It contains our x values, but we **must add a column of 1s** to the left. Why? To account for the bias term (w_0).

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \quad y = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}$$

Step 2: Compute $X^T X$

We multiply the transpose of X by X itself.

$$X^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1+1+1 & 1+2+3 \\ 1+2+3 & 1+4+9 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

Step 3: Compute the Inverse $(X^T X)^{-1}$

We need the inverse of that 2×2 matrix. Formula for inverse of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $\frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

- **Determinant:** $(3 \times 14) - (6 \times 6) = 42 - 36 = 6$.
- **Inverse:**

$$\frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} = \begin{bmatrix} \frac{14}{6} & -1 \\ -1 & \frac{3}{6} \end{bmatrix} = \begin{bmatrix} \frac{7}{3} & -1 \\ -1 & \frac{1}{2} \end{bmatrix}$$

Step 4: Compute $X^T y$

Multiply X transposed by the target vector y .

$$X^T y = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} (1 \cdot 2) + (1 \cdot 4) + (1 \cdot 5) \\ (1 \cdot 2) + (2 \cdot 4) + (3 \cdot 5) \end{bmatrix} = \begin{bmatrix} 11 \\ 25 \end{bmatrix}$$

Step 5: Final Weights (w^*)

Multiply the Result of Step 3 by the Result of Step 4.

$$w^* = \begin{bmatrix} \frac{7}{3} & -1 \\ -1 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 11 \\ 25 \end{bmatrix}$$

Calculations:

- $w_0 = (\frac{7}{3} \times 11) + (-1 \times 25) = \frac{77}{3} - \frac{25}{1} = \frac{2}{3}$
- $w_1 = (-1 \times 11) + (\frac{1}{2} \times 25) = -11 + 12.5 = 1.5 = \frac{3}{2}$

Solution: $w_0 = 0.67$, $w_1 = 1.5$. **Equation:** $\hat{y} = 1.5x + 0.67$

Part 3: Your Turn (Interactive Exercises)

Here are two new problems based on the same logic. Grab a piece of paper and try to set up the matrices!

Practice Problem 1 (Simple)

Dataset: You have 3 points: $(1, 1)$, $(2, 3)$, $(3, 4)$. **Goal:** Find w_0 (bias) and w_1 (slope).

Short Solving Explanation:

1. **Identify Matrices:**

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$$

2. **Compute $X^T X$:** (This is identical to the solved example because the input x 's are the same!)

$$(X^T X)^{-1} = \begin{bmatrix} \frac{7}{3} & -1 \\ -1 & \frac{1}{2} \end{bmatrix}$$

3. **Compute $X^T y$:**

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 + 3 + 4 \\ 1 + 6 + 12 \end{bmatrix} = \begin{bmatrix} 8 \\ 19 \end{bmatrix}$$

4. **Solve w^* :**

- $w_0 = \left(\frac{7}{3} \cdot 8\right) + (-1 \cdot 19) = \frac{56}{3} - \frac{57}{3} = -\frac{1}{3}$
- $w_1 = (-1 \cdot 8) + \left(\frac{1}{2} \cdot 19\right) = -8 + 9.5 = 1.5$

Answer: $w_0 = -0.33, w_1 = 1.5$. Line: $\hat{y} = 1.5x - 0.33$.

Practice Problem 2 (Clean Numbers)

Dataset: You have 3 points: $(0, 1), (1, 2), (2, 3)$. **Goal:** Find w_0 and w_1 .

Short Solving Explanation:

1. Identify Matrices:

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

2. Compute $X^T X$:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$$

3. Compute Inverse $(X^T X)^{-1}$:

- Determinant = $(3 \cdot 5) - (3 \cdot 3) = 15 - 9 = 6$.
- Inverse = $\frac{1}{6} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}$.

4. Compute $X^T y$:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 + 2 + 3 \\ 0 + 2 + 6 \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

5. Solve w^* :

$$\frac{1}{6} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

- $w_0 = \frac{1}{6}((5 \cdot 6) + (-3 \cdot 8)) = \frac{1}{6}(30 - 24) = \frac{6}{6} = 1$
- $w_1 = \frac{1}{6}((-3 \cdot 6) + (3 \cdot 8)) = \frac{1}{6}(-18 + 24) = \frac{6}{6} = 1$

Answer: $w_0 = 1, w_1 = 1$. Line: $\hat{y} = 1x + 1$. (This makes sense, look at the points: as x goes up by 1, y goes up by 1!).